The Motivation for Changes in the ArchiMate® 4 Specification

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Executive Summary

This document provides an overview of the motivation behind the most significant changes included in the ArchiMate® 4 Specification. It serves as a companion to the standard itself.

The aim of this new version is to stay true to the goals behind the ArchiMate Specification and carry out the necessary housekeeping to clean up and streamline the standard, to improve its ease of adoption and use. This helps to ensure that the complex needs of advanced users are not prioritized at the expense of those regular, average users who actually form the majority of users, as well as prospective users, whose concerns are often not heard. As such, this document supports The Open Group vision of Boundaryless Information Flow™.

Motivation for Changes

This document provides an overview of the motivation behind the most significant changes included in the ArchiMate® 4 Specification [C260]. It serves as a companion to the standard itself.

The difficulty with evolving standards like the ArchiMate Specification is that the people working on them are typically advanced experts who like to have advanced features for their advanced use cases. However, this gets in the way of regular, average users who are hindered by the resulting complexity. Moreover, the concerns of prospective users are often not heard at all, since they are not represented in standardization fora. There is often a focus on maintaining backward compatibility, caused by current users reluctant to change their ways of working or update their tools – but this leaves no room to accommodate the needs of those prospective users. Together, these factors often lead to inertia and bloat in standards.

This is why the "ArchiMate Manifesto" was formulated years ago, to focus on the proverbial "80%" of users and use cases:

- Simplicity over comprehensiveness

  The most important design consideration is that the language has been explicitly designed to be as small as possible, but still usable for most Enterprise Architecture modeling tasks. Many other languages try to accommodate the needs of all possible users. In the interest of simplicity in learning and use, the ArchiMate language has been limited to the concepts that suffice for modeling the "80%" of practical cases.

- Needs of the many over wants of the few

  Any change to the language shall cover a clearly identified and important use case of a substantial part of its users. In line with the previous principle, keeping things simple for the "80%" is more important than catering for the wants of the "20%".

- Collaboration over coverage

  If a need is already covered sufficiently by another modeling language, it is preferred to create a mapping to that language rather than incorporate those features into the ArchiMate Specification; for example, where a detailed software design model is better written in Unified Modeling Language™ (UML®), the ArchiMate application component concept can be used to map between both models.

- People over tools

  Understandability is key. The language was designed for communication among human users rather than for technical usage. There are many more technical languages already and, in line with the previous principle, the ArchiMate Specification should work together with those rather than try to cover their scope and level of detail too.

- Communicating architecture over other use cases

  The ArchiMate language was designed first and foremost to support communicating (enterprise) architecture. Accommodating other use cases should not get in the way of this key use case.

The strategy behind the ArchiMate 4 Specification is exactly this: Stay true to the goals behind the ArchiMate Specification and do the necessary housekeeping to clean up and streamline the standard. Of course, not everyone will like it if their favorite concept is taken away, even though there is always a substitute or conversion given in Version 4. It will indeed require some conversion and change both in ways of working and in tools, but in the end this is a clear improvement for the large majority of users. Most importantly, this will lower the barrier to entry for prospective users, who are sometimes deterred by the size and complexity of the language. Conversely, without such housekeeping, all legacy and "conceptual debt" from the past would be dragged along, burdening users for all eternity with ever more baggage. Such bloat has been the death of many other approaches in the past, and the intention is to consciously avoid that.

Merging Behavior Elements in the ArchiMate Core Metamodel

Importantly, the layers of the ArchiMate framework are not abstraction layers, where the Business Layer is a more abstract representation of the Application Layer, which in turn is an abstraction of the Technology Layer. The ArchiMate Business Layer is intended for modeling the human perspective on the architecture of an enterprise; i.e., what the people do, as opposed to what is implemented in some technology elements. It is defined by the standard as depicting "business services offered to customers, which are realized in the organization by business processes performed by business actors" [C226, §3.3], and it also states: "Business actors may be individual persons (e.g., customers or employees), but also groups of people (organization units) and resources that have a (or at least long-term) status within the organizations. Typical examples of the latter are a department and a business unit." [C226, §8.2].

So the intent of the Business Layer is explicitly not to be a kind of conceptual-level abstraction that is realized by the Application and Technology Layers below. Even though the framework structure is used by some as if it represents abstraction layers, that does not match its intent or the allowed relationships. This is most apparent when looking at the active structure elements in each layer, which are clearly not abstractions from lower layers; e.g., a business actor (e.g., a person or organizational entity) is not somehow realized by an application component. Abstraction in the ArchiMate 4 Specification [C260, §3.5] is dealt with not via layering but by using relationships (in particular, realization, assignment, and composition/aggregation).

Moreover, the current layered setup leads to limitations and extra work, even when it is used in this "conceptual" way. First of all, the current set of behavior elements is basically identical across the Business, Application, and Technology Layers. The only difference is who or what can perform this behavior; i.e., which active structure elements can be assigned to it. There is no inherent difference between, for example, a business process, an application process, and a technology process; semantically, all three model (causal/temporal) sequences of behavior. For some new users of the language, this triplication of concepts is puzzling and feels like an unnecessary overhead.

If we look at the main process modeling standard Business Process Model and Notation™ [BPMN™], we do not see such a layered distinction either. Its basic behavior concept of task can be specialized in various ways; e.g., as a manual task, send task, user task, service task, etc., but this specialization is not in any way layered. Moreover, it is the swimlane (representing the so-called participant) in which a task is put that determines who or what resource performs it; resources can be assigned flexibly, since no type restrictions apply.

Analysis of BPMN usage across multiple implementations indicates that it is typically used for describing administrative processes that consist of a combination of manual, IT-assisted, and fully-automated tasks, all in a single process. There is no natural way to do the same in an ArchiMate model, and for users familiar with BPMN, the ArchiMate behavior concepts therefore seem unnecessarily restrictive and complicated to use. The same is true for UML, where concepts such as action used in activity diagrams are not subdivided in categories according to who or what is supposed to perform them.

Another input to this is the North Atlantic Treaty Organization (NATO) Architecture Framework [NAFv4]. NATO has adopted the ArchiMate language as one of two approved metamodels for implementing NAFv4. If we look at its behavior concepts, we see generic concepts such as service, service function, and service operation, which had to be mapped to ArchiMate 3.2 layer-specific concepts, depending on the context.

Second, to properly model that some business process is fully or partially automated, create an application process with a realization relationship to that business process, and then assign an application component to it. This effectively duplicates part of the model without adding any insights or value. Moreover, the decision of whether some behavior should be manual or automated cannot be postponed to a later stage of the architecture and design process. A decision must be made straight away by making it a business, application, or technology process. Even when the current business process concept is (mis)used as a "conceptual" or "abstract" process as mentioned above, only business structure elements can be assigned to it, and it must be duplicated as described if it needs to be modeled an automated process. This leads to models like the one in Figure 1, where business process "BP 2" and the realization relationship is just overhead.

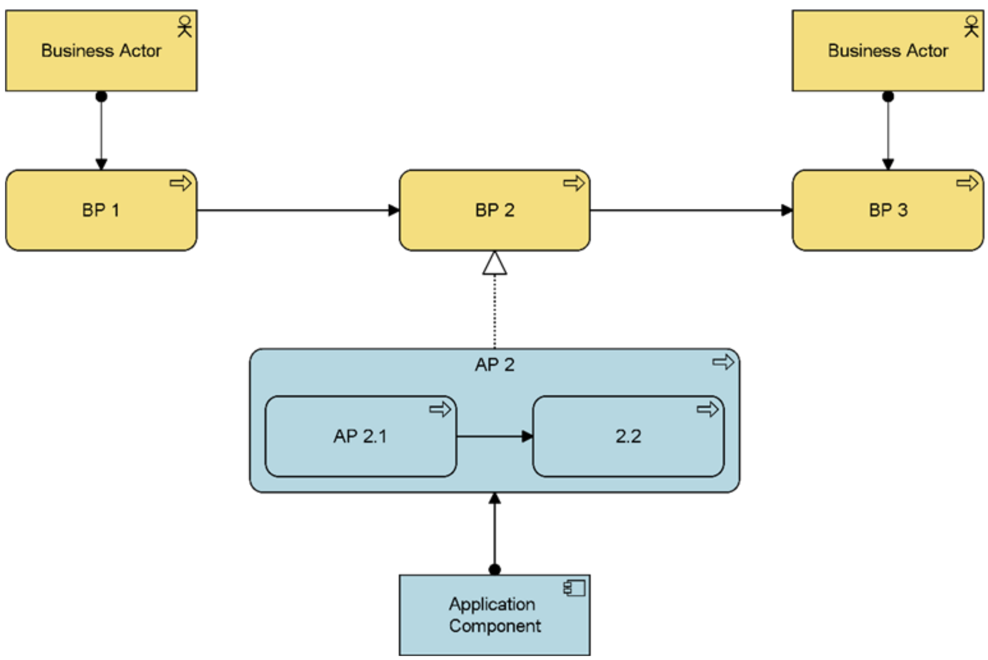

Figure 1: Modeling a Partially-Automated Process in Archimate 3.2 Specification

This construct, with additional elements and a realization relationship between the layers, was added as a work-around in Version 3 of the language. In previous versions, it was not possible to even express that some business process was completely or partially automated, since business processes were (and are) intended as "what people do": it could only be expressed that some application behavior served, but not realized, business behavior. This was in keeping with the intent that the Business Layer is not an abstraction of the Application Layer (nor is the Application Layer an abstraction of the Technology Layer). This work-around with realization relationships may have given that impression, but it would have been better had this problem been solved in a more fundamental way.

Third, it is currently not possible to define a process that is performed by (a collaboration of) active structure elements from different layers; e.g., a factory worker plus a robot, or an office worker plus an Artificial

Intelligence (AI) assistant. There is no such collaboration concept, nor is it allowed to assign active structure elements from multiple layers to one and the same behavior element. This kind of modeling pattern is becoming increasingly common, but the ArchiMate Specification cannot represent it in an easy way.

Solution

The update to the standard collapses the three sets of behavior concepts into a single one. So instead of, for example, business process, application process, and technology process, there is a single concept: "process". The assignment of an active structure element to that behavior element decides how this is executed. *De facto*, the generic behavior elements of ArchiMate 3.2 Specification [C226, §4.1.2], which are abstract and cannot be used to create concrete models, replaced the layer-specific variants of these.

The process modeled in Figure 1 can now be expressed in a simpler fashion, as shown in Figure 2.

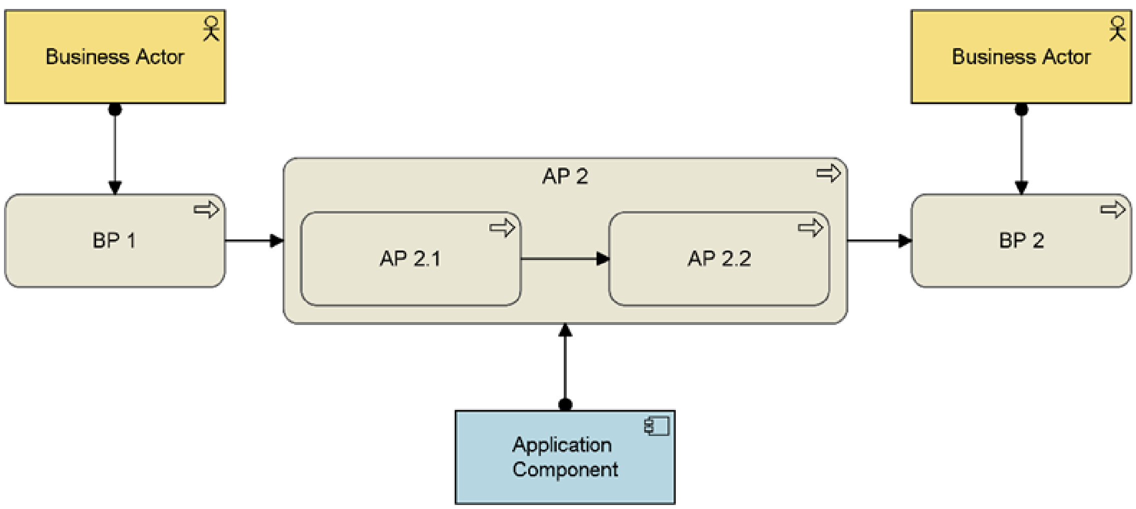

Figure 2: Modeling a Partially-Automated Process in the ArchiMate 4 Specification, using a Generic Process Element

Similarly, a single collaboration concept replaces the layer-specific collaboration concepts, and may aggregate active structure elements from all core layers. Next to this, a generic role concept replaces business role and may have assignment relationships from internal active structure elements from all core layers, not just the Business Layer.

Advantages

- The number of concepts in the language is reduced, facilitating the understanding of the language by new users
- The size and complexity of the resulting models is reduced, as shown by Figure 1 and Figure 2, which improves understanding
- Mapping between an ArchiMate process model and a (possibly more detailed) BPMN process model or UML activity diagram becomes more natural, and BPMN or UML users can more rapidly understand ArchiMate models
- The decision about how a behavior should be implemented (e.g., manual, automated, or a combination thereof) can be taken later in the architecture and design process, offering better support for initial, high-level modeling
- The new structure allows users to model collaborative execution of behavior across layers; e.g., where a combination of "human and machine" performs some process, an example of which is shown in Figure 3
- It also permits the assignment of non-human actors to roles, which is becoming increasingly relevant in the age of AI

  Note that this does not imply machines should become (legally) accountable. Roles are used to model the responsibility to perform certain tasks, but responsibility and accountability are different notions. If needed, a specialization of the role concept could be used to model accountability, perhaps limiting the assignment of such a role to only business actors.

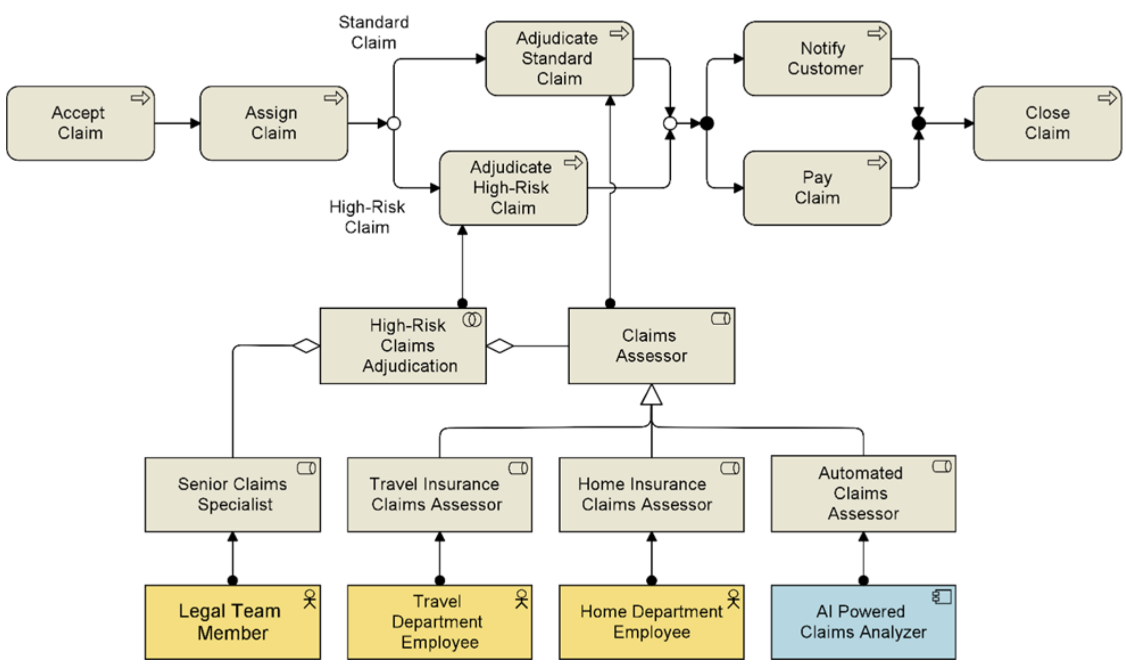

Figure 3: Example of Cross-Domain Use of Concepts

Figure 3 shows something that could only be modeled with difficulty in the past: An automated, AI-powered application for analyzing insurance claims that collaborates with human employees by fulfilling a role in the claim handling process. In the example, it would be allowed that this automated claims assessor adjudicates standard claims, but not high-risk claims, which always require the involvement of a senior specialist from the legal team.

In summary, the meaning of a model consisting of, for instance, a business actor assigned to a business process is exactly the same as that of a business actor assigned to a generic process; some actor performs some work. The only difference is that the first model contains a redundancy, in that it specifies in two ways that the behavior being performed is business behavior – both via the specific business process concept and also via the assignment of a business actor to it. So no meaning gets lost in merging these behavior concepts into a generic set, and we gain more flexibility in the design process and less complexity in the resulting models.

Conversion

To convert earlier concepts to the new structure, there are two main options.

The first option is to simply turn each layer-specific concept into the corresponding new generic concept, and change the realization from the application to the business process to a specialization or an aggregation relationship. Based on assignment relationships to those behavior elements, it is possible to deduce from which layer in the converted model they originated, except for behavior elements that do not have any assignments. It would be reasonable to assume that those are typically business processes, as in the case of BP 2 in Figure 4 – which illustrates the next figure for the example above.

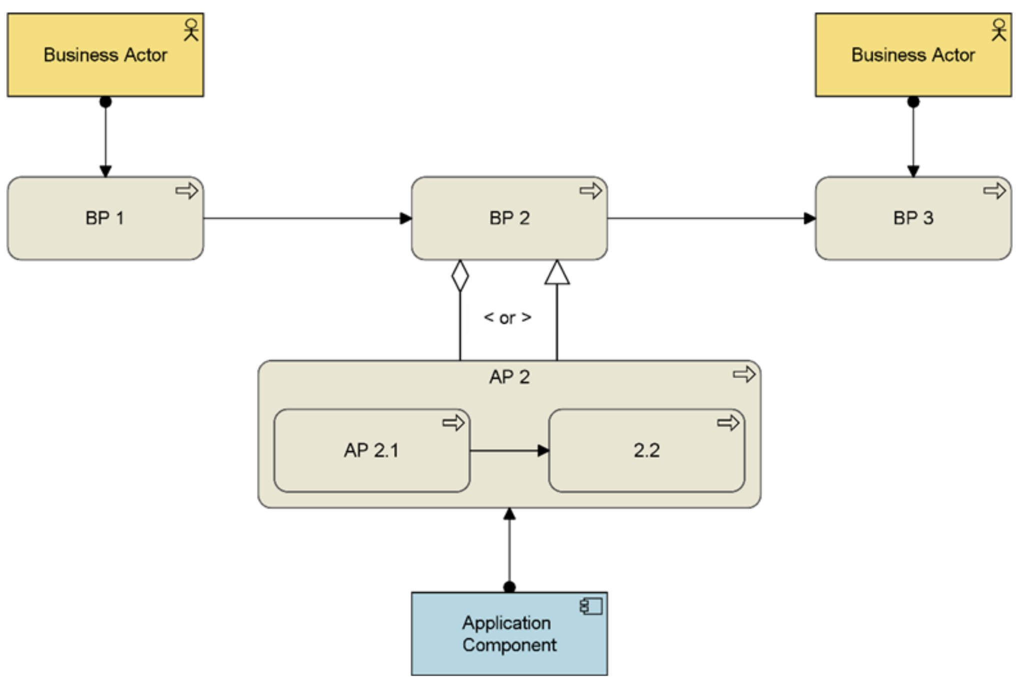

Figure 4: Converting an ArchiMate 3.2 Model without Specialization Profiles

The second option is to convert each layer-specific concept as above, and in addition assign a specialization profile [C260, §14, "Language"; §14.2, "Specialization of Concepts"] to show from which layer in the originating ArchiMate 3.2 model a new generic concept was converted. This way, no information is lost in the conversion.

This does require support for such concepts in the modeling tool used. The result is shown in Figure 5 (only showing the conversion of the realization to a specialization relationship here).

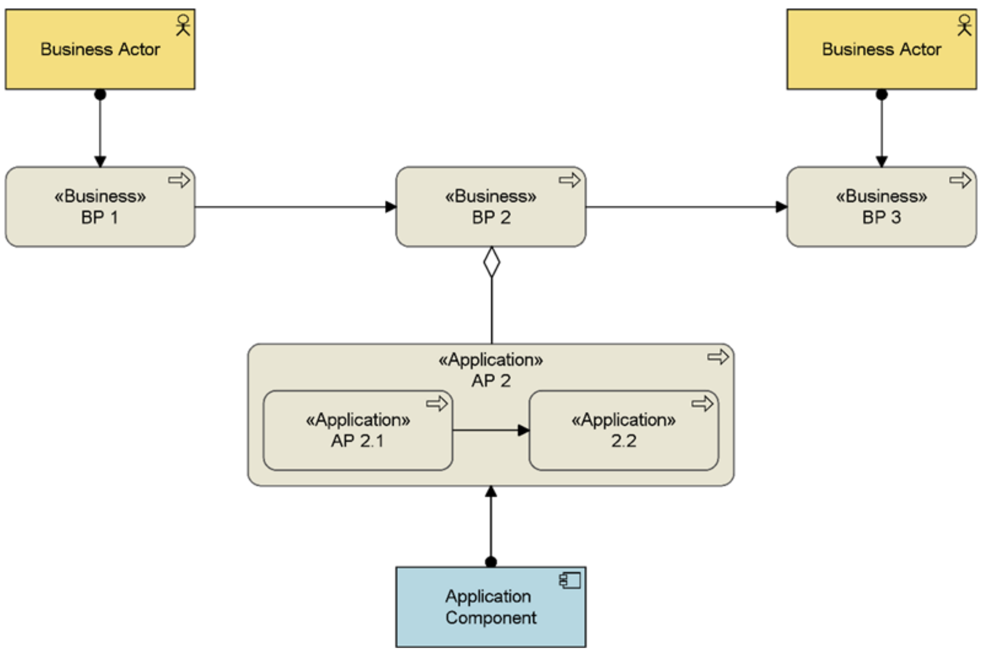

Figure 5: Converting an ArchiMate 3.2 Model with Specialization Profiles

A future version of the *ArchiMate® Model Exchange File Format for the ArchiMate Modeling Language* [C19C] is expected to add support exchanging specialization profiles.

Metamodel

The relevant portion of the core metamodel will then look as shown in Figure 6.

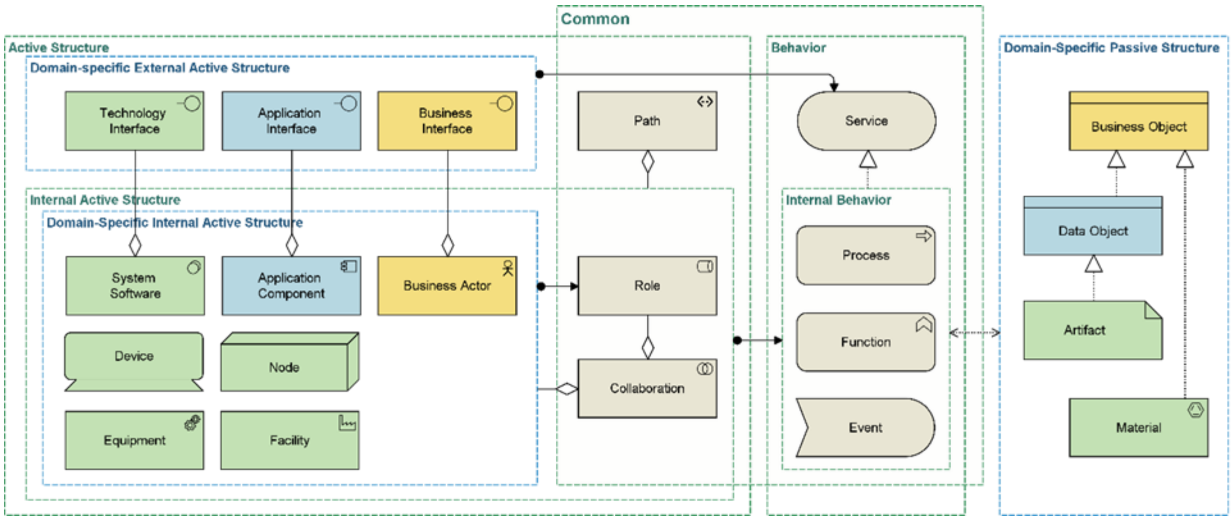

Figure 6: Metamodel Structure

Note that this does not show all connected concepts, only the ones where the new structure of common concepts has a direct impact.

New Language Structure Depiction

This new structure of the language also leads to a new hexagonal visualization. The new figure intentionally gets away from the layered matrix of previous versions. This is not just cosmetic: The notion of layers suggested a hierarchy, where lower layers are only there to serve or realize upper layers. Version 3.0 of the standard [C162] already relaxed the strict condition on serving relationships, that they should only point "upwards" between layers, but the framework image and the term "layer" were kept, still incorrectly suggesting such a hierarchy.

With the new image, we want to emphasize that there is no hierarchy implied by the structure of the language. Consequently, the term "layer", with its hierarchical connotations, is replaced by "domain". The TOGAF® Standard [C220] uses "architecture domain" in much the same way and distinguishes business, data, application, and technology as domains. The ArchiMate language matches the Business and Technology Domains, and combines the TOGAF application and data domains in a single Application Domain; moreover, it defines the Common, Motivation, Strategy, and Implementation and Migration Domains.

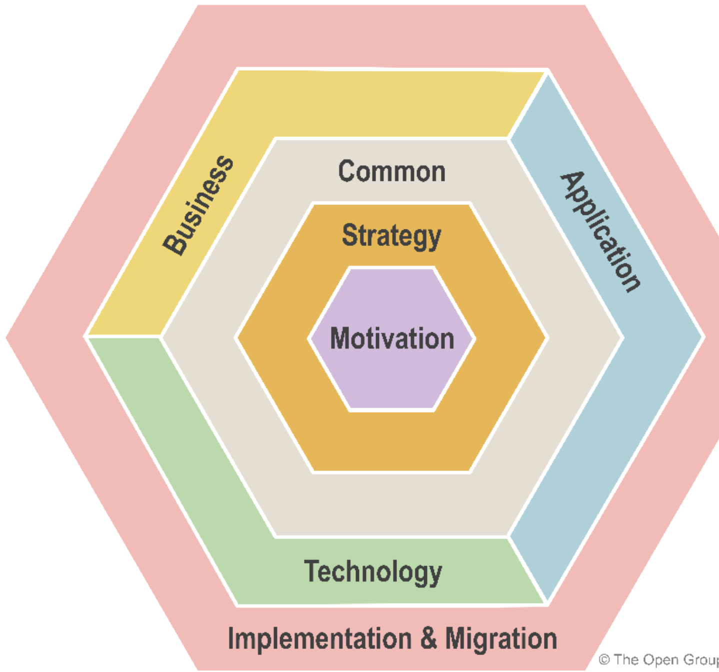

Figure 7: ArchiMate Hexagonion

Multiplicity on Relationships

The most important new feature is the option to specify the multiplicity of relationships. This takes basically the same form as in UML; i.e., n..m with a lower and upper bound, also allowing, for example, 0..* for an unbounded upper limit.

This was motivated mainly by Data Architects needing to model in more detail what the structure and constraints of their data models are. That is not the only use case of this feature though; it also allows the modeling of things like multiple process instances started from a single triggering event, a specific number of servers (instances of a server type) being part of a cluster, express that a relationship is compulsory like every service must be realized by at least one process, etc.

Deprecated Concepts

Next to merging and generalizing certain concepts, the ArchiMate 4 Specification also deprecates several concepts. The general thought behind this is to clean up the language and remove what is "clutter" for most users – distractions that increase the cognitive load of learning the language, which add complexity without sufficient benefits in return. In all cases though, a conversion is suggested that retains all information from models in ArchiMate 3.2 Specification, so nothing will be lost.

Interactions

Unlike other behavior elements, which are merged into a common set across the different architecture domains in the ArchiMate 4 Specification, the business, application, and technology interaction concepts are deprecated.

First of all, it turns out that these are hardly used in practice. An analysis (see Appendix A for a survey of approximately a dozen Bizzdesign® customers) of the frequency of use of all ArchiMate concepts shows that the three Interactions are all in the bottom part of that usage list (business actor, application component, and business process not surprisingly being the most popular concepts).

The meaning of Interactions is not very clear or understood; possibly the reason that most users ignore them. Are they sequence-like, akin to processes; i.e., performing some activities in a specific order? Or are they more function-like, collecting behavior using other criteria? If we consider them to be similar to Interactions in UML, they are indeed sequence-like, as can be seen in UML's Sequence diagram (one of its two Interaction diagrams); those sequences of activities are clearly a kind of process in the ArchiMate Specification, since that is defined as a sequence of behaviors.

If Interactions are indeed like the process concept, their only differentiating property is that they must be performed by more than one active structure element. With the introduction of multiplicities [C260, §5.6] of the snapshot), this can now be modeled in a much simpler and more flexible and expressive way, by using a cardinality such as '2..*' to express that at least two active structure elements (e.g., roles, business actors, etc.) must have an assignment relationship with some process.

For conversion from the ArchiMate 3.2 Specification, the process concept can be used, with possibly a specialization profile like in the other cases mentioned.

Representation

The representation concept is a left-over from the very first version of the ArchiMate Specification, when we wanted to express, for example, paper forms and letters that represent a business object, so we could model the ways in which communication is implemented. It then started to be used for electronic means as well (e.g., email messages, PDF files, etc.), where data object would have been a better choice. In the ArchiMate 3.0 Specification, the material concept was added, which can be used to model the physical representations of information on paper, to model, for example, the logistics of postal mail or the use of paper forms as a fallback scenario when the IT fails. And the more general case is to use business object (or a specialization thereof) instead.

Contract

In the ArchiMate metamodel, contract was already a specialization of business object. Its only real use is in the context of the product concept, where it is used to stipulate the agreement between the provider and consumer of that product. This can be equally well modeled with a regular business object, with a specialization profile if needed (e.g., when converting from ArchiMate 3.2 Specification).

Constraint

This is one more concept that was already defined as a specialization of another one in the metamodel, in this case of requirement. Semantically, it is often not very clear what the difference is, since both represent some boundary on the design or implementation space. Is applying, say, some legal regulation like General Data Protection Regulation (GDPR) best modeled as a constraint or as a requirement on your architecture?

Moreover, various other kinds of specializations of requirement appear in practice; e.g., to capture notions like "feature", "user story", "use case", "business rule", etc. Predefining one such specialization, and a fairly unclear one at that, is not very helpful. Again, using specialization profiles allows the addition of those specializations of requirement needed for use cases, without overburdening the language and all its users with them. Such a specialization can of course also be used when it is needed to convert your models from the ArchiMate 3.2 Specification and want to keep this distinction.

Gap

Gap has always been a rather odd concept. It represents a "statement of difference between two plateaus", according to its definition. However, with those two plateaus, that difference in the model can already be seen and analyzed by checking their relationships with other elements. Its lack of purpose also shows in its frequency of use, again in the bottom 10% of the analysis mentioned.

If this really needs to be represented by a separate element in the model, the assessment concept (or a specialization thereof) will do, since this can model any "analysis of the state of affairs of the enterprise" as its definition says, which easily accommodates the outcome of some gap analysis. Alternatively, (a specialization of) the deliverable concept can be used, if gap was used to represent the document coming out of a gap analysis activity (itself modeled as a work package), rather than the analysis result itself.

Why Not Simplify More?

Some may also question why there was not more simplification. In particular, the Technology Domain may still look somewhat crowded (although the path concept has been moved to the Common Domain). However, there are several reasons why it looks like this:

- The Technology Domain comprises two rather different kinds of technology: IT and physical (bits *versus* atoms)

  Had those two been colored differently, the picture would look more balanced, but since they are often combined in a single piece of technology (e.g., computer-controlled physical machinery), it was decided to keep them in one domain.

- The distinction between software and hardware is quite fundamental in IT, as is that between processing and networking

  Reducing the active structure elements to just node would be simpler, but too restrictive for many use cases. Still, in high-level models it might be enough to just use nodes.

- The physical technology concepts are somewhat underused (see also the statistics in Appendix A), because most Enterprise Architects are only concerned with the IT world and not with physical tech

  They are important for specific use cases though, and there is a growing use of these concepts. One example is NATO, who would not have adopted the ArchiMate Specification without these concepts (their "enterprise" of course being rather physical in nature). Other examples include operational resilience analysis (see, for example, the recent European Union (EU) Digital Operational Resilience Act (DORA) regulation), where to model data centers, office buildings, power supplies, etc., and Industry 4.0 (e.g., robotization, digital twins).

- Technology Architects often have a rather concrete view of the world and already find the ArchiMate Specification quite abstract: Infrastructure architects are often asking for more, not fewer concepts, because they want to model different kinds of devices (servers, switches, routers, etc.); cloud architects would like to have all their various AWS™/GCP™/Microsoft® Azure® elements, etc.

  So it is important to strike a balance here.

Principles for Definitions

The following principles are used to enable good definition in the ArchiMate Specification:

- Do not use a word which is the name of a concept if there is no intention to refer to this concept
- Prefer "concrete" names (e.g., interface) over "abstract" ones (e.g., external active structure element)
- Describe what the concept is, not its relationships with other concepts
- A definition should be understandable "stand-alone" (without having to know in advance the whole metamodel)

This has led to some changes in the definition of concepts, although not in their actual meaning.

Appendix A: ArchiMate Concept Usage Statistics

The data used here was collected across 17 medium to large user organizations in a wide variety of sectors, including government, banking, insurance, consumer goods, food, infrastructure, energy, and manufacturing. These organizations together represent several thousand users of the ArchiMate Specification, including many hundred active modelers (and others who are consuming the content). In total, this sample includes over 600,000 elements and nearly 1,000,000 relationships.

Chart 1 and Chart 2 both show the absolute number of instances of each concept (relationships and elements) per organization. Note that both charts use a logarithmic axis, to accommodate the large variance in the data, so the length of each colored sub-bar is not a linear representation of the corresponding organization's sample size.

Chart 3 shows the frequency of these concepts relative to model size, averaged over all organizations, again on a logarithmic axis. This shows, for instance, that on average, business actors make up a little over 15% of the elements in the sampled models, and application components nearly the same percentage. This average frequency provides a somewhat more balanced metric, because it avoids that one organization using many instances of a specific concept will dominate the result. In the end though, the order of elements sorted by relative frequency does not differ very significantly from the order sorted by absolute usage; the most significant change is location being ranked lower, because one of the organizations uses a very detailed hierarchy of locations, from continents via countries, cities, and sites down to buildings and rooms. This average frequency averages out such anomalies.

Note that of course frequency of use is not the only measure of a concept's relevance; e.g., stakeholder and principle are important concepts, but organizations will typically not model thousands of these.

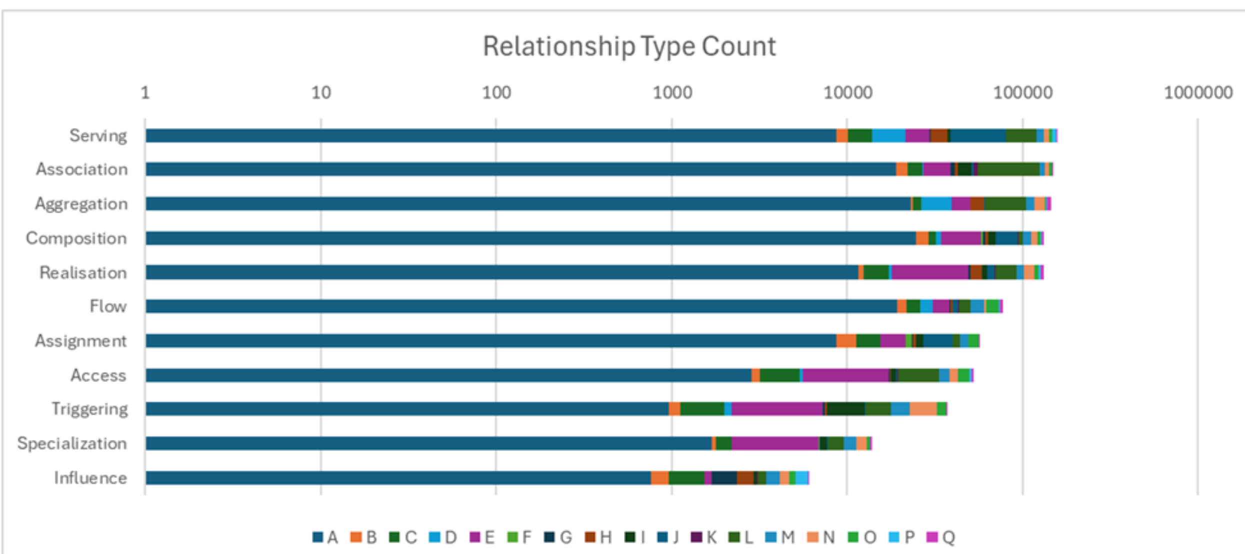

Chart 1: Absolute Number of Relationships

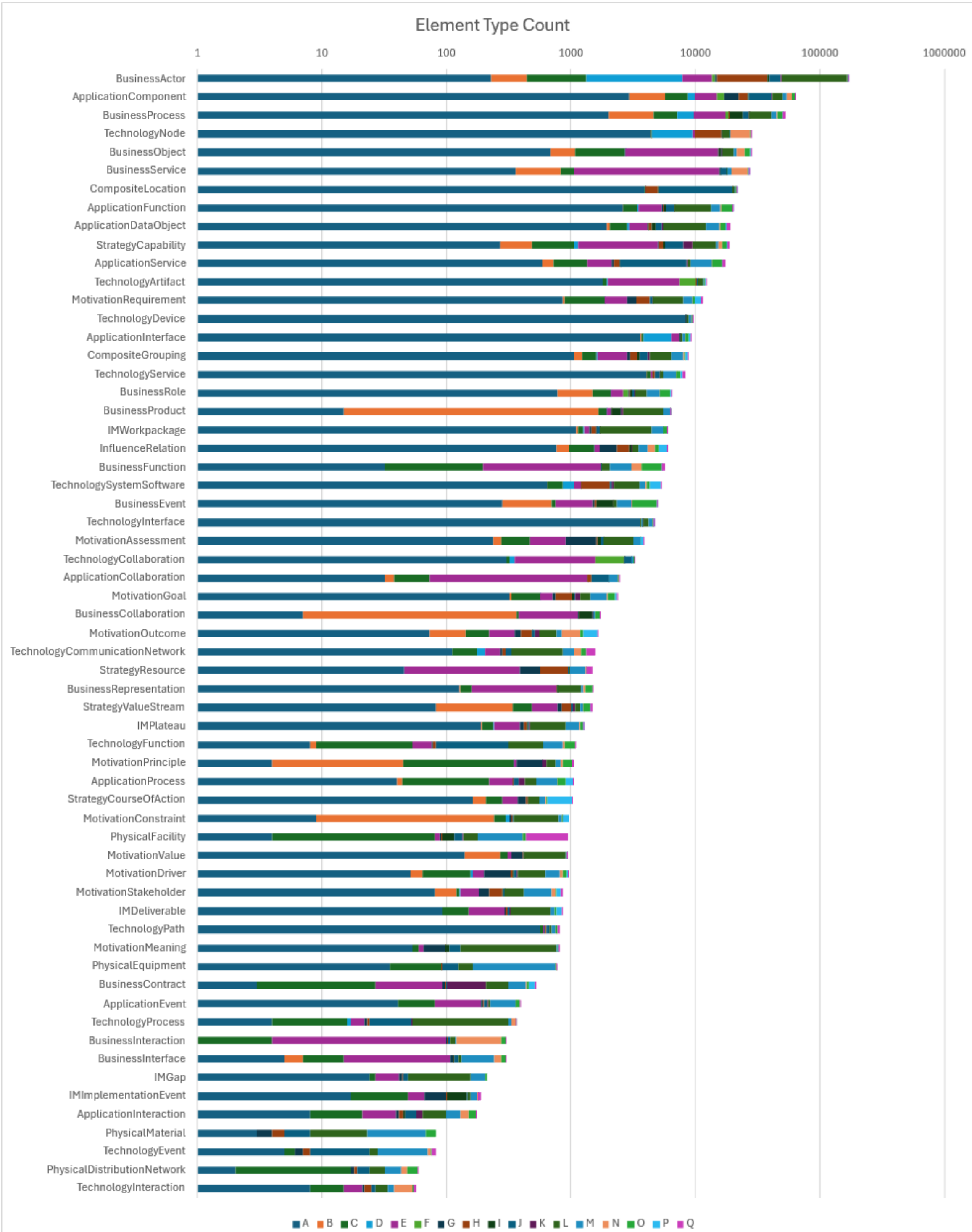

Chart 2: Absolute Number of Elements

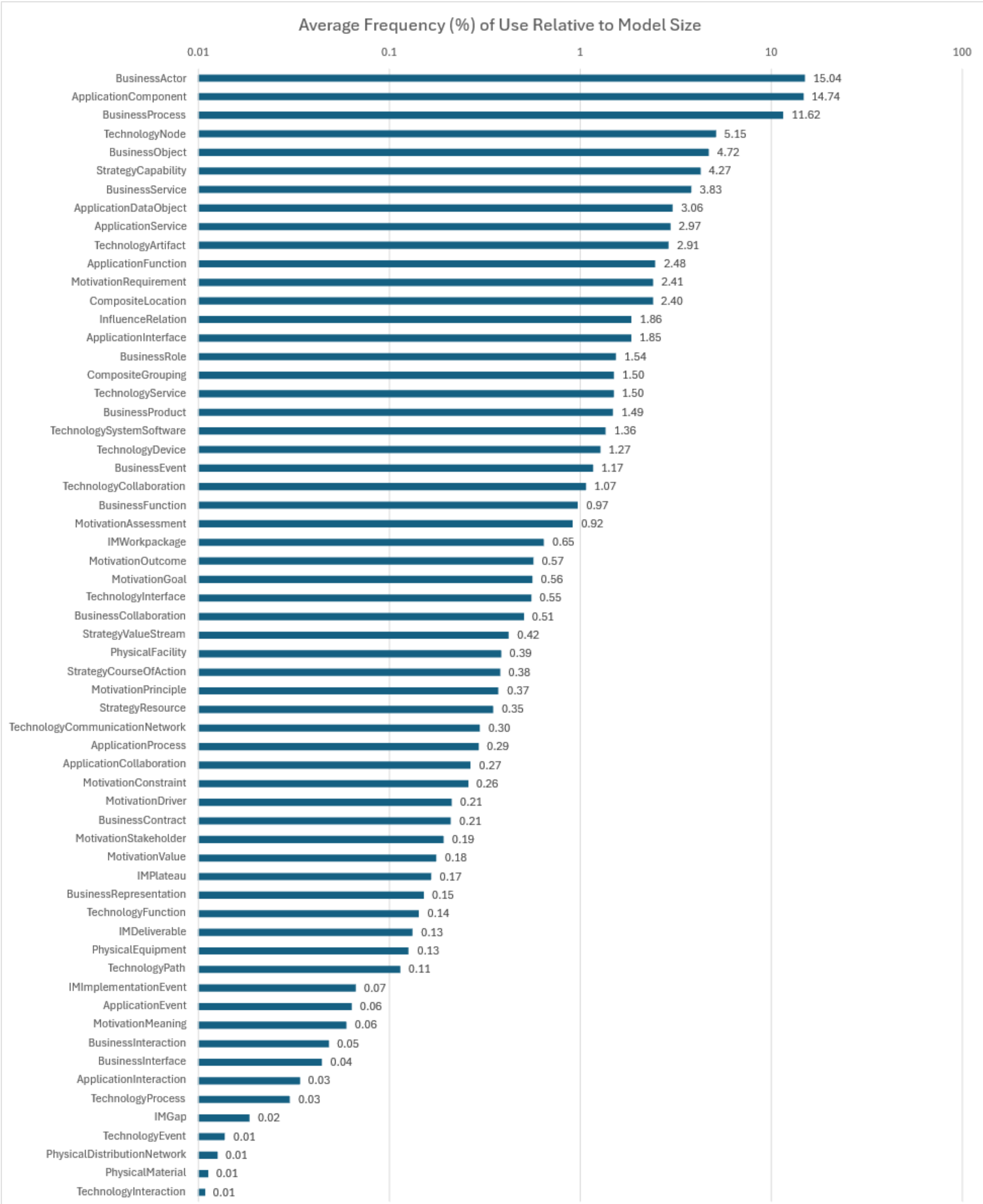

Chart 3: Frequency of Use Relative to Model Size

Referenced Documents

(Please note that the links below are good at the time of writing but cannot be guaranteed for the future.)

[BPMN] Business Process Model and Notation (BPMN) Specification, published by Object Management Group® (OMG®); refer to: www.omg.org/bpmn/#:~:text=The%20Business%20Process%20Model%20and%20Notation%20(BPMN),between%20business%20process%20design%20and%20process%20implementation**

[C162] ArchiMate® 3.0 Specification, a standard of The Open Group (C162), published by The Open Group, June 2016; refer to: www.opengroup.org/library/c162

[C19C] ArchiMate® Model Exchange File Format for the ArchiMate Modeling Language, Version 3.1, a standard of The Open Group (C19C), published by The Open Group, November 2019; refer to: www.opengroup.org/library/c19c

[C220] TOGAF® Standard, 10th Edition, a standard of The Open Group (C220), published by The Open Group, April 2022 and updated May 2025; refer to www.opengroup.org/library/c220

[C226] ArchiMate® 3.2 Specification, a standard of The Open Group (C226), published by The Open Group, October 2022; refer to: www.opengroup.org/library/c226

[C260] ArchiMate® 4 Specification, a standard of The Open Group (C260), published by The Open Group, April 2026; refer to: www.opengroup.org/library/c260

[NAFv4] NATO Architecture Framework, Version 4, published November 2025; refer to: www.nato.int/content/dam/nato/webready/documents/publications-and-reports/2504-NAFv4-ArchiMate.pdf

About the Authors

Marc Lankhorst

As Managing Consultant and Chief Technology Evangelist at Bizzdesign, Marc Lankhorst contributes to Bizzdesign's vision, market development, consulting, and coaching on digital business design and Enterprise Architecture. He has managed the development and evolution of the ArchiMate® language for Enterprise Architecture modeling since its very beginning. Based on this experience, he helps organizations with implementing and using the ArchiMate Specification in practice.

Jean-Baptiste Sarrodie, BNP PARIBAS

Jean-Baptiste Sarrodie has been working in the field of IT as a manager and an architect in several industries for over 20 years. He now works as an Enterprise Architect for BNP PARIBAS bank and divides his time between accompanying the bank's transformation projects and working on the evolution of the ArchiMate® Specification within The Open Group.

Since 2013, Jean-Baptiste has been contributing to the development of the open source modeling tool, Archi. Jean-Baptiste is also Chair of The Open Group ArchiMate Forum.

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- Working with suppliers, consortia, and standards bodies to develop consensus and facilitate interoperability, to evolve and integrate specifications and open source technologies
- Offering a comprehensive set of services to enhance the operational efficiency of consortia
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